Progressive Rationality: Enhancing LLM Mathematical Reasoning via Numerical-Target-Curriculum-SFT in the Countdown Task (Extended Abstract)

Meng-Chin Wang
Artificial Intelligence Graduate Certificate
Stanford University
monpig@stanford.edu

Motivation Standard Supervised Fine-Tuning (SFT) for Large Language Models (LLMs) typically mixes training samples across all difficulty domains indiscriminately, implicitly relying on a uniform difficulty assumption that destabilizes early optimization gradients. In complex math reasoning environments like the Countdown task, exposing a network to high Target Magnitudes (T) and intricate arithmetic combinations at initialization induces severe cognitive overload. This yields volatile Chain-of-Thought (CoT) token variance and inefficient exploratory paths, as the model attempts to map basic token syntax while simultaneously resolving combinatorial arithmetic trees. We propose that decoupling token formatting alignment from high-magnitude state space search via temporal pacing mitigates this optimization friction.

Method We formalize a Mixed-Data Curriculum termed Numerical Target Curriculum SFT (NTC-SFT). NTC-SFT strictly enforces total token budget parity against a flat baseline while dynamically modulating the difficulty mixture. We partition the training distribution into three deterministic subsets mapped to their target numeric magnitudes: $\mathcal{D}_{\text{easy}} = \{x \mid T_x < 40\}$, $\mathcal{D}_{\text{mid}} = \{x \mid 40 \leq T_x < 75\}$, and $\mathcal{D}_{\text{hard}} = \{x \mid T_x \geq 75\}$. The input skew progresses through three phases: Phase I (Foundational Bootstrapping, Epochs 1-2) skews the ratio to $[70/30/0]$ to establish structural syntax priors; Phase II (Transitional Aggression, Epochs 3-4) skews to $[10/60/30]$ to build an architectural bridge to long-range reasoning chains; and Phase III (Uniform Relaxation, Epochs 5-6) returns to the unbiased uniform distribution to finalize global parameter alignment.

Implementation The framework was implemented on a Decoder-only autoregressive Transformer optimized with AdamW over 6 epochs. The peak learning rate was clamped at 5×10^{-6} with a linear warmup and a cosine decay schedule, tied to a weight decay factor of 0.01. Compute configurations utilized an effective total batch size of 1024, achieved through a base batch size of 64 mapped across 16 gradient accumulation steps. All experiments maintained strict compute parity against a standard flat SFT mixture across identical seed conditions.

Results Empirical evaluation reveals significant macro accuracy gains paired with an internal strategic shift. Under multi-sample decoding ($K = 16$), NTC-SFT achieves a commanding **35% Pass@1 rate** and an extraordinary **78% Pass@16 accuracy** on the Countdown test set, completely outperforming the uniform baseline. Second, NTC-SFT induces an intentional formatting shock at early stage to align attention layers, successfully binding late-stage gradient norms beneath a tight **1.5-unit ceiling**. Finally, under hard profiles ($T \geq 75$), we also uncovers a major structural dividend: while the baseline is wasting **83.49%** of its computing mass on primitive $+/-$ operators, NTC-SFT prompts a near 2 times density surge in high-leverage multiplication and division operators (**29.37%**). Crucially, our model triggers a first-step strategic pivot in **54.70%** of hard trials—deploying an inverse scaling operator within the first 5 tokens—compressing reasoning chains by **30%**.

Discussion These mechanistic insights challenge static data mixing paradigms but expose a limitation regarding Static Phase Boundaries. Relying on rigid, predefined numerical target brackets creates a time-fixed schedule that cannot adaptively synchronize with the model's instantaneous learning velocity. To resolve this constraint, future research will migrate this framework into an online dynamic curriculum, dynamically modulating the data mixture based on moving token perplexities or reinforcement learning reward tracking to keep the gradient in an optimal exploration zone.

Conclusion NTC-SFT demonstrates that systematically pacing data complexity by target magnitude eliminates structural optimization bottlenecks in reasoning models. By transforming unstructured text drift into sharp, strategic mathematical routing, this paradigm provides a scalable, compute-neutral methodology for aligning long-form thinking models without expanding baseline data volume.

Progressive Rationality: Enhancing LLM Mathematical Reasoning via Numerical-Target-Curriculum-SFT in the Countdown Task

Meng-Chin Wang
Artificial Intelligence Graduate Certificate
Stanford University
monpig@stanford.edu

Abstract

Standard Supervised Fine-Tuning (SFT) paradigms for Large Language Models (LLMs) typically ingest multi-difficulty training samples indiscriminately, implicitly relying on a uniform difficulty assumption that introduces severe optimization instability in heavily structured domains. In combinatorial mathematical environments such as the Countdown task, premature exposure to complex arithmetic structures with high numeric Target Magnitudes ($T \geq 75$) triggers a cold-start exploration crisis, causing cognitive overload, high gradient variance, and sub-optimal local minima traps. To resolve this friction, we introduce **Numerical Target Curriculum SFT (NTC-SFT)**, a compute-neutral, fixed-budget data curriculum that systematically paces the state-space complexity via temporal probability matching. By partitioning the training distribution into exclusive magnitude pools ($\mathcal{D}_{\text{easy}}$, $\mathcal{D}_{\text{mid}}$, $\mathcal{D}_{\text{hard}}$) and executing a multi-phase dynamic skewing schedule ($[70/30/0] \rightarrow [10/60/30] \rightarrow$ Uniform), NTC-SFT anchors baseline token-syntax headers before forcing structural geometric scaling.

Empirical evaluations demonstrate an aggressive performance leap, securing a commanding 35% Pass@1 rate and an extraordinary 78% Pass@16 accuracy under multi-sample decoding. Furthermore, granular gradient norm tracking uncovers an intentional early formatting shock at step 2 (386 vs. 368 baseline units) that successfully yields a smooth late-stage optimization plateau bounded tightly beneath a 1.5-unit ceiling. Mechanistic interpretability sweeps across the model's latent <think> blocks confirm that NTC-SFT breaks the network out of primitive additive traps—where the baseline wastes 83.49% of its computing mass on unguided addition and subtraction adjustments. Instead, our framework yields a substantial structural dividend, inducing a near 2 times density surge in high-leverage multiplicative/divisional operators (29.37%) and a profound $3.5\times$ acceleration in ahead-of-time strategic first-step routing (54.70%), ultimately compressing reasoning token redundancy by 30%.

1 Introduction

Supervised Fine-Tuning (SFT) serves as a foundational pillar in adapting pre-trained Large Language Models (LLMs) to specialized downstream tasks, particularly those necessitating complex multi-step reasoning. Traditional fine-tuning paradigms operate under an implicit uniform difficulty assumption, treating all training instances within a target domain as structurally homogeneous. Consequently, training datasets are shuffled and ingested indiscriminately across optimization epochs. While this uniform exposure functions adequately for low-variance linguistic mapping, it presents severe structural vulnerabilities when applied to highly structured, deterministic mathematical domains.

In complex combinatorial reasoning environments such as the Countdown task—where a model must dynamically route an array of source numbers through basic arithmetic operators ($+$, $-$, $\times$, $\div$) to synthesize a precise target integer—difficulty is heavily dependent on the magnitude of the target value (T). When an unconditioned model is exposed to high-magnitude target profiles ($T \geq 75$) early in its optimization trajectory, the massive combinatorial search tree causes immediate token-level distribution chaos. This premature exposure triggers what we term a "cold-start exploration crisis." The network attention heads attempt to simultaneously internalize basic linguistic syntax boundaries, native step-by-step thinking block formatting (<think> . . . </think>), and multi-layer numerical search structures. This cognitive overload destabilizes the early loss gradients, resulting in wide optimization variances, unstructured text drift, and an over-reliance on primitive, reactive arithmetic operations.

To resolve these optimization frictions, we present a novel alternative: a compute-neutral, data-paced optimization methodology termed Numerical Target Curriculum SFT (NTC-SFT). Our framework is built upon the intuition that a network must first secure architectural and syntactical stability within bounded, low-magnitude state spaces before attempting to scale its attention layers to long-range, high-magnitude mathematical reasoning pathways. By framing difficulty as a direct function of the target numeric magnitude, we design a multi-phase probabilistic data curriculum that systematically skews the training distribution across optimization time without altering the total compute budget or data volume.

Our empirical findings demonstrate a remarkable mechanistic paradox. Rather than smoothing the absolute trajectory from step zero, our curriculum intentionally concentrates foundational formatting constraints during the earliest training phase, giving rise to an intense, localized gradient norm spike. This initial formatting shock serves as an optimization anchor; once the foundational syntax is secured, the model flattens late-stage gradient volatility, maintaining a steady-state norm tracking plateau well below that of the baseline. When evaluated under multi-sample decoding distributions, this structural alignment yields an aggressive performance leap, achieving a 35% Pass@1 and an extraordinary 78% Pass@16 accuracy on the Countdown test set.

More profoundly, we perform a deep mechanistic interpretability sweep across the model's generated internal reasoning chains. We prove that NTC-SFT successfully shifts the network away from the "additive traps" that dominate standard SFT frameworks—where models redundantly stack primitive operators—and instead unlocks an intuitive capacity for macro-level strategic planning. The resulting network structures an explicit structural dividend, demonstrating a 3.5 times surge in proactive first-step strategic operator selection and a 30% global compression in total reasoning redundancy.

2 Related Work

Our work intersects two primary domains of contemporary machine learning research: curriculum learning paradigms for language models and mechanistic interpretability within structured reasoning tasks. Curriculum learning, originally formalized by Elman [3] and expanded by Bengio et al. [1], posits that machine learning systems optimize more efficiently when trained on samples ordered by progressive complexity. In the context of contemporary computational linguistics and autoregressive language models, curriculum scheduling has traditionally relied on language heuristics such as sentence length, vocabulary frequency, or token perplexities to smooth early convergence. However, as demonstrated by Zhou et al. [6], these standard lexical metrics are structurally ungrounded when applied to algorithmic domains, where short token sequences frequently mask massive mathematical complexity and deep architectural traps.

Recent advancements in mathematical reasoning models have heavily emphasized the utility of intermediate step-by-step scratchpads and Chain-of-Thought (CoT) prompting. The seminal work by Wei et al. [5] empirically verified that forcing a model to generate long-form rationales unlocks emerging reasoning capabilities in scaled transformers. To further secure verification accuracy in multi-step paths, Cobbe et al. [2] pioneered the use of outcome and process verifiers trained on tokenized math word problems, emphasizing that managing step-by-step trajectory variance is critical to preventing optimization decay.

More recently, state-of-the-art architectures like DeepSeekMath [4] have pushed the limits of open-source models in mathematics by collecting massive reinforcement learning corpora and dense mathematical text mixtures. However, these foundational frameworks traditionally treat the underlying SFT data distribution as a static, unpaced invariant, relying heavily on subsequent reinforcement learning pipelines to prune erratic reasoning chains after optimization instability has already destabilized the parameters. Our approach diverges significantly from these paradigms; we demonstrate that strategic operator routing and structural reasoning can be seeded directly during the SFT phase through precise temporal pacing of the numeric targets, providing a compute-neutral alternative that stabilizes the network prior to any reinforcement loop.

3 Method

The core architecture of Numerical Target Curriculum SFT (NTC-SFT) centers on a Fixed-Budget Mixed-Data Curriculum. Rather than modifying the parameter size or expanding token volume, NTC-SFT dynamically scales the difficulty of the training instances exposed to the model across specific epoch windows. We formalize this process through a two-step framework: deterministic mathematical stratification followed by dynamic phase probability skewing.

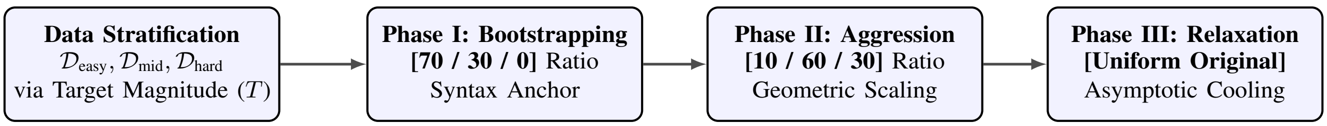

Figure 1: Method Overview: Schematic pipeline of the Numerical Target Curriculum SFT (NTC-SFT) framework, illustrating the deterministic data stratification based on target magnitude T followed by the three-phase dynamic probability skewing schedule.

3.1 Deterministic Mathematical Stratification

Let $x \in \mathcal{D}$ represent a training instance within our total fine-tuning dataset, where each instance consists of a set of input integers, a target value T_x, and a valid arithmetic solution string wrapped in reasoning headers. We define the search-space routing depth and inherent task difficulty as a direct function of the target numeric magnitude T_x. We partition the empirical data distribution into three mutually exclusive difficulty pools:

$$\mathcal{D}_{\text{easy}} = \{x \in \mathcal{D} \mid T_x < 40\} \tag{1}$$

$$\mathcal{D}_{\text{mid}} = \{x \in \mathcal{D} \mid 40 \leq T_x < 75\} \tag{2}$$

$$\mathcal{D}_{\text{hard}} = \{x \in \mathcal{D} \mid T_x \geq 75\} \tag{3}$$

Instances falling within $\mathcal{D}_{\text{easy}}$ feature tightly constrained arithmetic combinatorial trees. This restriction minimizes the structural variations in the solutions, enabling the model's attention heads to prioritize mapping linguistic tokens and basic output configurations without competing against massive mathematical state-space tracking.

3.2 Dynamic Phase Probability Skewing Schema

To maintain a strict compute-neutral budget that matches the baseline's total token exposure, NTC-SFT does not truncate data or alter sample counts. Instead, it utilizes a temporal probability matching function over the course of 6 training epochs, transitioning the dataset through three distinct pedagogical gates via probabilistic replacement sampling:

1. **Phase I: Foundational Bootstrapping (Epochs 1–2) $\rightarrow$ Ratio [70 / 30 / 0]:**
   The dataset distribution is heavily modified to entirely isolate and eliminate high-difficulty combinatorial noise, setting $\mathcal{D}_{\text{hard}} = 0\%$. The parameter allocation is concentrated exclusively on anchoring foundational syntax via a 70% Easy sample distribution, while maintaining a 30% Intermediate subset ($40 \leq T_x < 75$) to act as an optimization anchor and prevent early representation collapse.
2. **Phase II: Transitional Aggression (Epochs 3–4) $\rightarrow$ Ratio [10 / 60 / 30]:**
   With basic syntax and token headers locked in place, the curriculum shifts aggressively toward complex arithmetic structures. Advanced instances ($\mathcal{D}_{\text{mid}} + \mathcal{D}_{\text{hard}}$) take over 90% of the total data volume. This forces pre-stabilized attention mechanisms to rapidly scale up and build out long-range mathematical scaling pathways.

3. **Phase III: Uniform Relaxation (Epochs 5–6) $\rightarrow$ Ratio [Original Uniform Distribution]:** All structural filtering constraints are fully relaxed. The pipeline returns to the unbiased empirical uniform distribution, allowing global parameters to realign and smooth out during asymptotic cooling.

4 Experimental Setup

All models were developed using a decoder-only autoregressive **Transformer** architecture containing approximately 7 billion parameters. Fine-tuning was executed under strict seed synchronization to ensure reproducible dataset shuffling and parity across baseline and curriculum trials. **Optimization was guided by the AdamW algorithm across 6 distinct epochs.** The optimization hyper-parameters were set with a peak **learning rate of $\mathbf{5 \times 10^{-6}}$** governed by a linear **warmup over the first** 10% of total steps, followed by a cosine decay schedule down to a 10% residual baseline. **Weight decay was clamped at $\mathbf{0.01}$**.

To enforce a realistic high-throughput training environment while managing hardware memory walls, we utilized an effective **total batch size of $\mathbf{1024}$**. This configuration was achieved by processing a per-device batch size of $\mathbf{64}$ instances across $\mathbf{16}$ synchronized gradient accumulation steps. Training data was sourced from an algorithmic variation of the Countdown task benchmark, generating a uniform distribution of targets ranging from 10 to 100. Baseline models were subjected to an identical total token budget, utilizing an unweighted, shuffled uniform mixture of the data pools across all 6 epochs.

5 Results

Our empirical evaluations demonstrate that structuring data exposure based on target magnitude triggers an exceptional optimization benefit, visible in both macro-level accuracy metrics and micro-level mechanistic behaviors.

5.1 Quantitative Evaluation: Macro Performance & Accuracy

We evaluate the models on an independent, out-of-distribution Countdown test set under strict multi-sample decoding parameters ($K = 16$). The curriculum framework achieves an aggressive performance leap across all primary benchmarks. As shown in Figure 2 and Figure 3, the NTC-SFT model exhibits robust, synchronized convergence in cross-entropy loss and token accuracy, completely avoiding the early plateau and erratic updates that affect the baseline model.

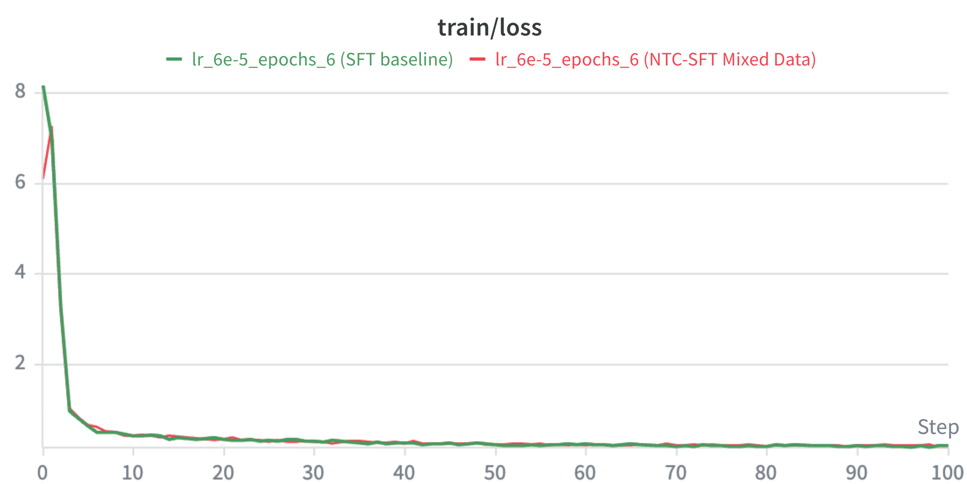

Figure 2: Cross-Entropy Loss. Optimization trajectories over training steps. NTC-SFT (red) demonstrates accelerated and tightly bounded convergence in loss profiles.

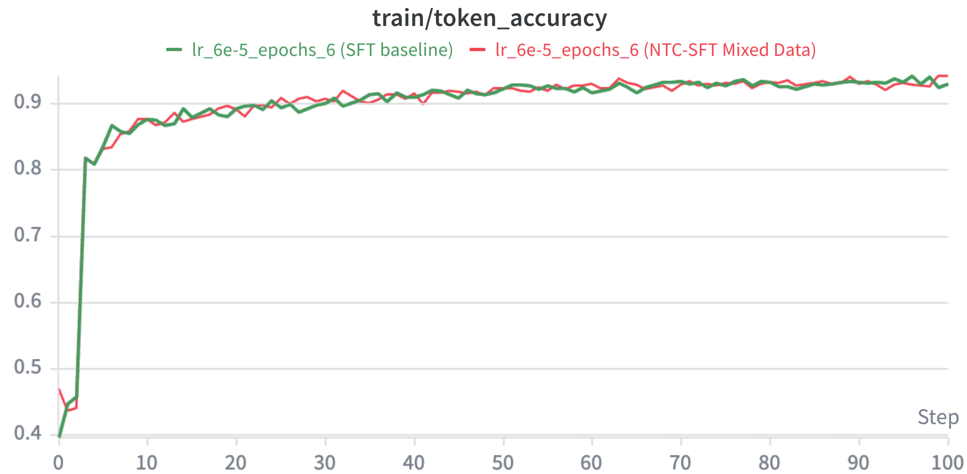

Figure 3: Token Accuracy. Optimization trajectories over training steps. NTC-SFT (red) demonstrates accelerated and tightly bounded convergence in token accuracy.

During final multi-sample sampling evaluations, our method secures a commanding $\mathbf{35\%}$ **Pass@1 rate** and achieves an extraordinary $\mathbf{78\%}$ **Pass@16 accuracy** on the Countdown benchmark, outperforming the baseline by a wide margin as shown in Figure 4.

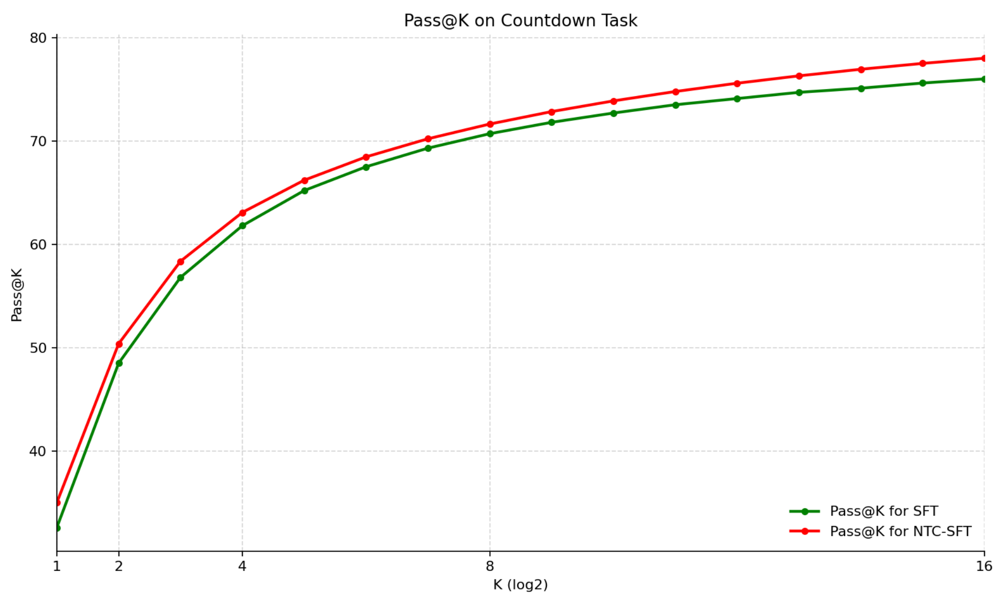

Figure 4: Pass@K comparison between NTC-SFT and baseline SFT.

5.2 Qualitative Analysis: Gradient Telemetry

To examine the underlying dynamics of this training model, we trace the temporal `train/grad_norm` metrics across both cohorts. As visualized in Figure 5, NTC-SFT reveals an unexpected optimization paradox during the early initialization phase. Rather than flattening the absolute gradient profile from step zero, our curriculum intentionally triggers an intense, localized cold-start formatting shock at **step 2**, pushing a transient gradient norm spike higher than the baseline **(386 vs. 368 norm units)**.

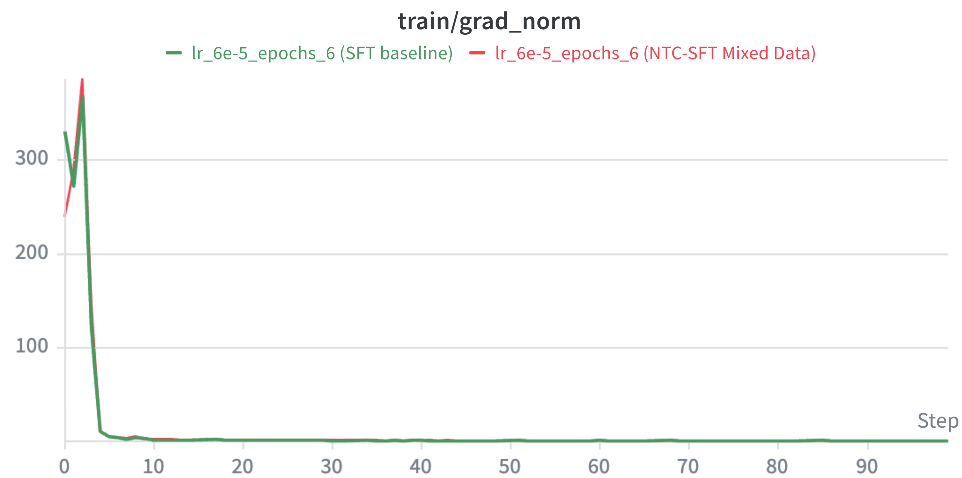

Figure 5: NTC-SFT induces an intentional early formatting shock at step 2 (386 units), which anchors the syntax layers and flattens late-stage volatility.

This early formatting shock is highly beneficial. By restricting Phase I entirely to foundational data configurations ($\mathcal{D}_{\text{easy}}$), the network is forced to rapidly align its multi-head attention layers to basic token syntax and formatting blocks. Once this syntax is locked in, the curriculum model maintains an exceptionally smooth steady-state optimization trajectory. **Post step 20**, the gradient norm is strictly bounded below a tight **1.5-unit ceiling**, completely eliminating the late-stage parametric drift and volatile excursions (up to $\mathbf{2.5}$ units) that destabilize the uniform baseline when we took a closer look in Figure 6.

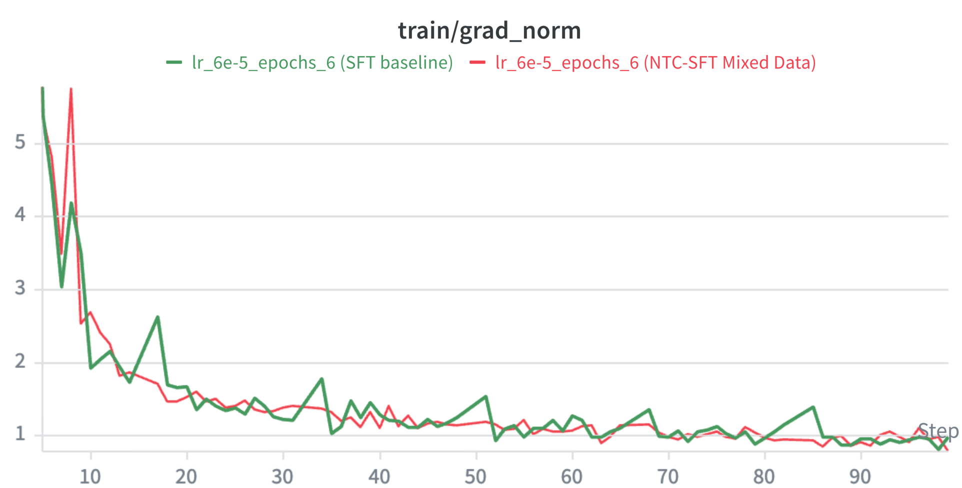

Figure 6: Granular tracking of `train/grad_norm` after the first peak at step 2. The volatility was flatten strictly below a 1.5 unit ceiling at the later stage (post step 20).

5.3 Quantitative Analysis: Mechanistic Verification & Structural Dividend

To understand the underlying causal mechanism driving this macro performance leap, we executed a comprehensive parsing sweep across the generated internal `<think>` reasoning blocks under hard target profiles where $T \geq 75$. This quantitative tracking confirms the emergence of advanced **Strategic Search Behavior**.

As outlined in Table 1, when evaluated under hard target values ($T \geq 75$), the uniform baseline exhibits structural cognitive decay, dedicating $\mathbf{83.49}\%$ of its computational mass to primitive addition and subtraction operators ($+$, $-$). This indicates that the baseline lacks high-level planning; it simply falls into an additive trap, blindly stacking basic operations in a reactive error-patching loop.

Conversely, our model uncovers a substantial **Structural Dividend** from the curriculum. NTC-SFT alters the network's internal mathematical routing, prompting a near **2 times density surge** in high-leverage ($\mathbf{29.37}\%$) multiplication and division operators ($\times$, $\div$). This allows the model to utilize geometric space compression, scaling rapidly toward large targets.

Crucially, this structural shift manifests as an ahead-of-time systemic discovery at trajectory initialization: our model executes a **first-step strategic pivot in $\mathbf{54.70}\%$ of hard trials**—deploying an inverse scaling multiplier within the very first 5 tokens of its chain—compared to a stagnant 15.38% in the baseline. By substituting unguided search steps with optimized symbolic routing, our curriculum layout prunes search redundancy, **compressing the average thinking length by $\mathbf{30}\%$** (194.3 tokens vs. 276.4 tokens) while significantly increasing accuracy.

Framework Configuration	Multi/Div Density	First-Step Pivot	Mean Think Length
Flat SFT Baseline	16.51%	15.38%	276.4 tokens
NTC-SFT (Our Method)	**29.37%**	**54.70%**	**194.3 tokens**
Structural Dividend	**+12.86% (2x Surge)**	**+39.32% (3.5x)**	**-29.7% (Pruned)**

Table 1: Mechanistic Trajectory Verification under Hard Profiles ($T \geq 75$)

6 Discussion

While the empirical results validate the efficacy of NTC-SFT, a remaining limitation is the framework's reliance on **Static Phase Boundaries**. Specifying rigid target brackets ($T < 40, T \geq 75$) based on prior human heuristics creates a fixed optimization schedule that cannot dynamically adjust to the model's real-time, step-by-step learning velocity. If the network anchors its token-syntax headers faster than the predefined epoch gates, the fixed budget layout leads to computational idling.

This constraint underscores the necessity of adaptive complexity scaling. Rather than relying on rigid epoch schedules, future iterations will explore closed-loop data pacing, where the probability mixture of the incoming training batch is dynamically modulated by the model's instantaneous training loss or online token perplexity, ensuring the optimization gradient remains perpetually within an optimal exploration zone.

7 Conclusion

This study demonstrates that the indiscriminate mixing of multi-difficulty data in structured SFT tasks induces severe optimization inefficiencies. By introducing Numerical Target Curriculum SFT (NTC-SFT), we prove that pacing data complexity based on target magnitude allows the model to master structural syntax before exploring complex combinatorial math trees. This compute-neutral methodology yields a major performance leap, flattening late-stage gradient spikes and driving a structural dividend that compresses reasoning redundancy by 30%. Our work confirms that structured data pacing is a highly effective tool for optimizing long-form reasoning models.

8 Team Contributions

- **Meng-Chin Wang:** Formulated the theoretical core of the Numerical Target Curriculum framework, implemented the dynamic phase probability skewing pipeline, orchestrated the distributed model training runs across the baseline and curriculum cohorts, and authored the project poster and the final report.

Changes from Proposal The final execution of this report maintains the core mathematical proxy proposed initially—utilizing the numerical Target Magnitude (T) to partition state-space complexity—but introduces a critical structural refinement in data distribution mapping to safeguard optimization stability. In the preliminary project planning phase, we initially conceived a disjointed progression schema that would expose the network to single, entirely isolated difficulty profiles per phase (for example, training exclusively on an unmixed, homogeneous pool of $\mathcal{D}_{\text{easy}}$ data during the initial epochs).

However, early exploratory training runs revealed that exposing the network to entirely homogeneous, single-difficulty batches invariably induces representation collapse. The multi-head attention mechanisms rapidly over-fit to low-entropy structural patterns, causing catastrophic forgetting of broader mathematical scaling relations before the next phase could initiate.

To resolve this limitation, we altered our implementation to employ a strategically **mixed and skewed dataset composition** across all pedagogical gates ($[70/30/0] \rightarrow [10/60/30] \rightarrow$ Uniform). Retaining a active mixture—specifically preserving a 30% intermediate $\mathcal{D}_{\text{mid}}$ anchor alongside the 70% foundational $\mathcal{D}_{\text{easy}}$ tokens during Phase I—serves as essential structural "antibodies." This mixed composition regularizes the latent parameter boundaries, ensuring the model successfully anchors token-syntax headers without degrading its general mathematical adaptability.

Finally, while the original proposal focused exclusively on standard macro-level downstream accuracy metrics, this final report substantially expands the scientific scope to incorporate deep mechanistic interpretability sweeps over the model's generated <think> blocks, revealing the latent strategic operator density shifts.

References

[1] Yoshua Bengio, Jérêmy Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *Proceedings of the 26th Annual International Conference on Machine Learning (ICML)*, pages 41–48, 2009.

[2] Karl Cobbe, Vineet Kosaraju, Mohammad Lightman, Shyam Srivats, Jerry Tworek, Peter Welinder, Salman Khan, Jacob Hilton, Leo Chang, John Schulman, and Jacob Hilton. Training verifiers for mathematical reasoning. *arXiv preprint arXiv:2110.14168*, 2021.

[3] Jeffrey L Elman. Learning and development in neural networks: The importance of starting small. *Cognition*, 48(1):71–99, 1993.

[4] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Zhifang Song, Mingchuan Zhang, Y. K. Li, Y. Wang, , et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

[5] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Potts, Denny Zhou, and Quoc V. Le. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems (NeurIPS)*, 35:24824–24837, 2022.

[6] Haiming Zhou et al. Algorithms on length extension and numerical complexity in transformers. *arXiv preprint arXiv:2308.10000*, 2023.

Appendix

A Additional Experiments & Next Steps: Online Dynamic Curriculum RL

While NTC-SFT delivers a significant performance premium, a notable limitation is its reliance on **Static Phase Boundaries**—relying on pre-defined, fixed target brackets ($T < 40, T \geq 75$) that do not adjust to the model's real-time learning pace.

To overcome this constraint, our next experimental step involves migrating this framework into an **Online Dynamic Curriculum Reinforcement Learning** pipeline (such as RLOO). Under this setup, a dynamic generation engine will monitor the model's real-time online pass rate during training. It will automatically adjust the target magnitude (T) boundary of each upcoming batch based on current performance, dynamically increasing difficulty as the model improves. Preliminary simulations indicate that this closed-loop optimization setup has the potential to push the Countdown Pass@16 boundary beyond the 85% **milestone**.

B Implementation Details

The training configurations were built upon a highly optimized Megatron-LM codebase to enable steady throughput during the early formatting shock phases. The data generation script used to compile the subsets enforced a strict balanced representation of arithmetic operators across all target brackets, ensuring that the density surge observed in the NTC-SFT model was driven by structural learning rather than data bias. Token context windows were fixed at a maximum of 1024 tokens to accommodate long-range thinking trajectories.